Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/20/2004
PI Name:	Sung, Chih-Jen Ph.D.		
Project Title:	CATALYZED COMBUSTION IN MICRO-PROPULSION DEVICES		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cleveland	**State:**	OH
Zip Code:	44106-7204	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Note: Original end date was 11/30/2007.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schneider, Steven (NASA Glenn Research Center)		
Grant/Contract No.:	NNC04AA15A		
Performance Goal No.:			
Performance Goal Text:			
	The proposed program addresses the need to understand the physics and chemistry involved in propellant combustion processes in micro-scale combustors for propulsion systems and energy conversion devices on micro-spacecraft. These spacecraft are planned to have a mass less than 50 kilograms with attitude control estimated to be in the 10 milli-Newton thrust class. Combustors of this size are expected to benefit significantly from surface catalysis processes. Therefore, the proposed program specifically aims to gain a fundamental understanding of the ignition conditions, flammability limits, combustion efficiency, and modeling of propellant combustion in heated microtubes fabricated from known catalyst materials. The data and models generated will form a micro-combustor database, which is applicable to MEMS based chemical thrusters and energy conversion devices. The present study is broken down into the following four major projects: (1) experimental determination of the ignition and flammability limits as functions of Reynolds number,		

Task Description:	pressure, preheat temperature, catalyst type, and tube length to inside diameter ratio; (2) experimental mapping of wall temperature distribution and exit profiles of temperature and species; (3) experimental assessment of performance and efficiency for various micro-combustors; and (4) detailed modeling of catalytic combustion including gas phase kinetics and surface chemistry. Three different, readily available, catalyst materials, namely platinum, palladium, and nickel, will be evaluated. Gaseous propellant selections for this program include H2/O2, CO/O2, and CH4/O2. They are selected because their non-toxic character will aid in the human exploration of space. It is noted that the productions of the aforementioned fuel are near-term top priorities of in-situ resource utilization (ISRU) technologies. These proposed studies are anticipated to engender significant impacts on fundamental understanding and predictive capabilities regarding catalytic combustion in general and micro-propulsion devices in particular. Ultimately, the studies will provide insights into approaches for achieving increased energy conversion efficiencies and leading to technological advances in various combustion processes/devices in terrestrial, space-based, and extraterrestrial surroundings.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is relevant to the development of micro-satellites, which are important for Earth weather and resource reconnaissance. The present investigation can also improve the efficiency of combustion devices on Earth and the micro-scale devices developed will greatly aid the portability of power supplies for Earth applications.
Task Progress:	Micro-propulsion devices behave differently than macro-scale devices because of the differences in magnitude of flow rates and heat transfer. Reducing the combustor size increases the relative surface area, increasing the heat loss, and as combustors are continuously reduced in size, they approach the quenching dimensions of the propellants. Combustors of this size are expected to benefit significantly from surface catalysis processes. A miniature flame tube apparatus is chosen for this study because microtubes can be easily fabricated from known catalyst materials and their simplicity in geometry can be used in fundamental simulations for validation purposes. Experimentally, we investigated the role of catalytically active surfaces within 0.4 and 0.8 mm internal diameter microtubes, with special emphases on ignition processes in fuel rich gaseous hydrogen and gaseous oxygen. Calculations of flame thickness and reaction zone thickness predict that the diameters of our test apparatus are below the quenching diameter of the propellants in most atmospheric test conditions. The temperature and pressure rise in resistively heated platinum microtubes and the exit hydrogen concentration were used as an indication of exothermic reactions. Data on imposed heat flux/preheat temperature required to achieve ignition versus mass flow rate are presented. With a plug flow model, the experimental conditions were simulated with detailed gas-phase chemistry and surface kinetics. Computational results generally support the experimental findings.
Bibliography Type:	Description: (Last Updated: 12/29/2006)
Abstracts for Journals and Proceedings	Schneider, S.J.; Boyarko, G.A.; Sung, C.J. "Catalyzed Ignition of Bipropellants in Microtubes" Forty-First Aerospace Sciences Meeting AIAA Paper No. 2003-0674 , Jan-2003
Abstracts for Journals and Proceedings	Schneider, S.J.; Sung, C.J.; Boyarko, G.A. "Catalyzed Combustion of Bipropellants for Micro-Spacecraft Propulsion" Thirty-Ninth AIAA/ASME/SAE/ASEE Joint Propulsion Conference AIAA Paper No. 2003-4924 , Jul-2003
Abstracts for Journals and Proceedings	Boyarko, G.A.; Sung, C.J.; Schneider, S.J. "Catalyzed Combustion of Hydrogen-Oxygen in Platinum Tubes for Micro-Propulsion Applications" Forty-Second Aerospace Sciences Meeting AIAA Paper No. 2004-0303 , Jan-2004
Articles in Peer-reviewed Journals	Boyarko, G.A.; Sung, C.J.; Schneider, S.J. "Catalyzed Combustion of Hydrogen-Oxygen in Platinum Tubes for Micro-Propulsion Applications" Proceedings of the Combustion Institute, Volume 30 , Dec-2004
Awards	Schneider, S.J.; Sung, C.J.; Boyarko, G.A. "Northern Ohio AIAA Best Paper Award" May-2004
Dissertations and Theses	Boyarko, G.A. "Catalyzed Combustion of Hydrogen-Oxygen in Platinum and Palladium Tubes for Micro-propulsion Applications" Master Thesis, Case Western Reserve University May-2004
Presentation	Sung, C.J. "Micro-Chemical Propulsion: Combustion Challenges" Department of Mechanical Engineering, Southern Taiwan University of Technology, Tainan, Taiwan Dec-2003
Presentation	Sung, C.J. "Micro-Chemical Thrusters" Department of Vehicle Engineering, National Pingtung University of Science and Technology, Pingtung, Taiwan Dec-2003